

Water from Ground to Sky

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Plant and atmosphere

- Water is critical for agricultural production
- Plants use water to maintain their structural integrity, to move nutrients, photosynthate and hormones from places where they are produced to places where they are consumed.
- Water loss by plants is inevitable: water leaves through the same stomata that CO₂ enters.
- Plant roots take water from the soil, move it osmotically through the plant, lose it through stomata on the leaves through diffusion.

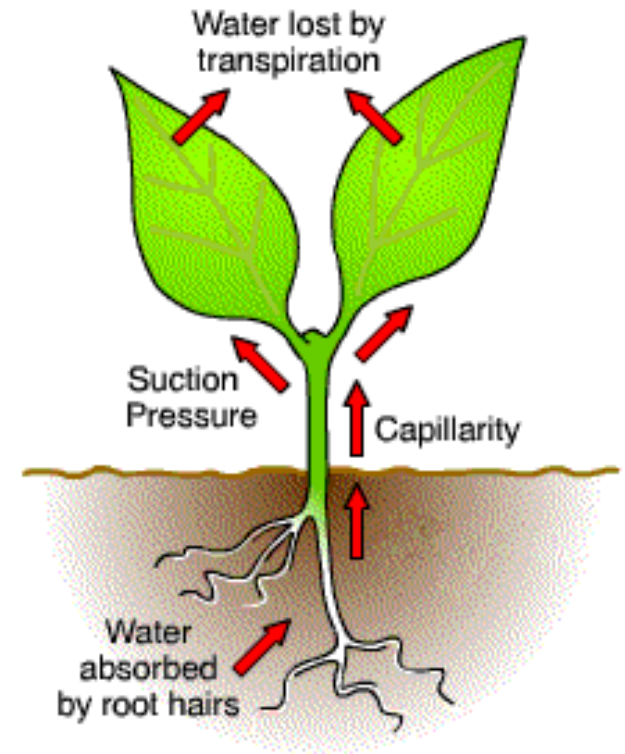
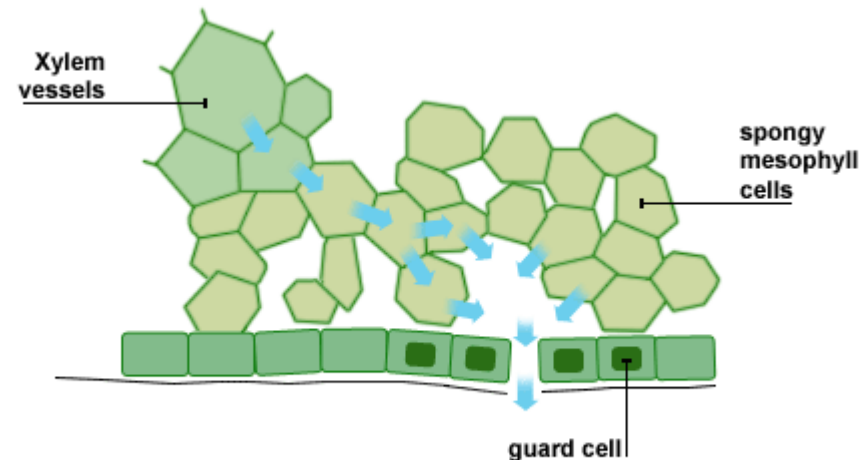
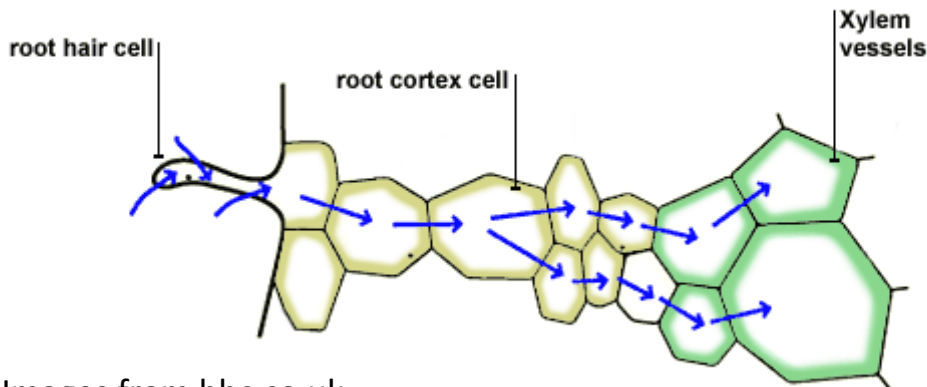


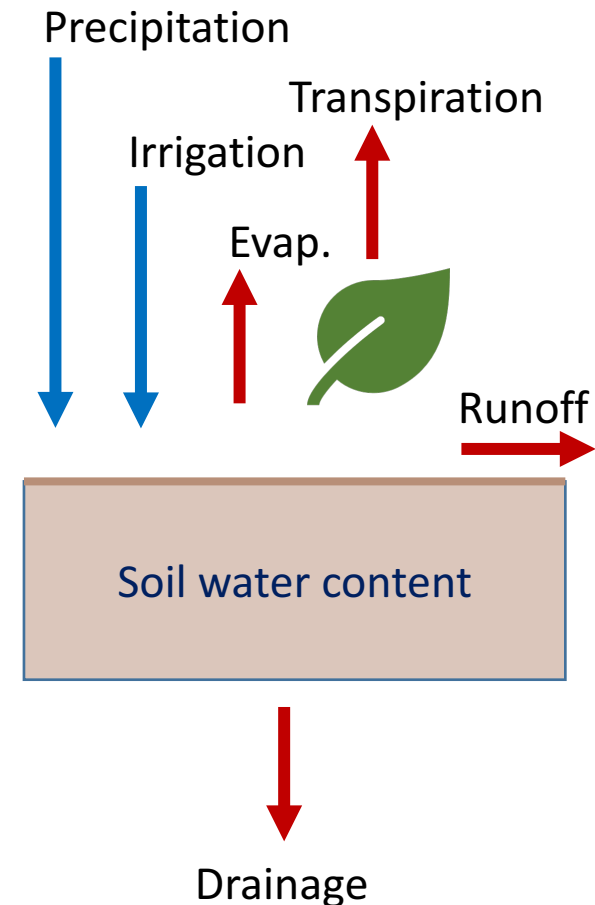
Image from wizznotes.com



Images from bbc.co.uk

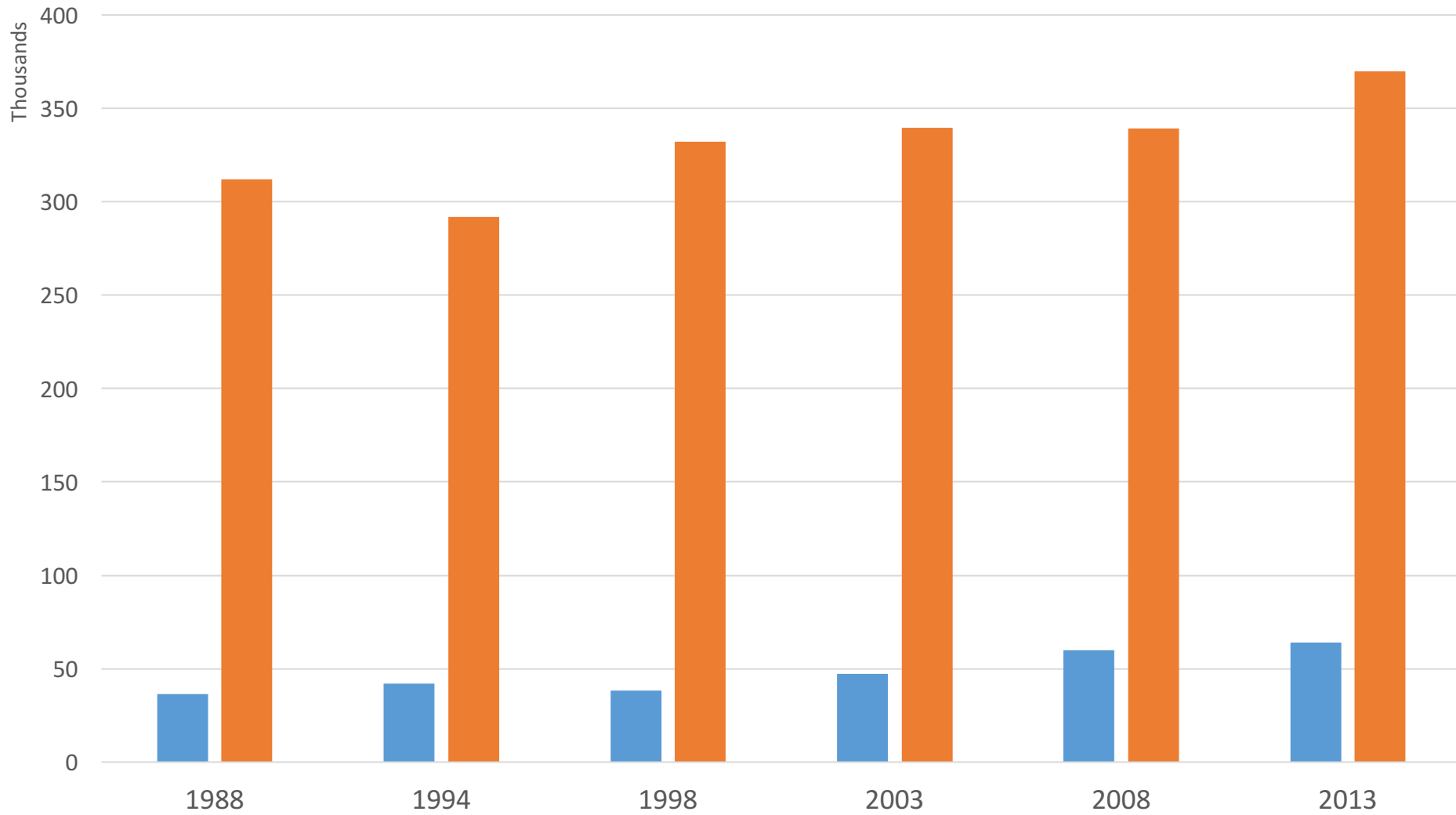
Soil Water Balance

- Challenge: Water is a finite resource: it is in increasingly short supply for agricultural use.
- There are ways to optimize its usage: too much water can be wasteful or even detrimental to the crop; too little creates a variety of problems: lower yields, greater susceptibility to diseases, etc.
- Principled methods of optimizing irrigation usage involve some kind of water balance calculation.
- Managing the water balance components for optimality: Scientific Irrigation Scheduling



SIS vs Non-SIS Methods Used

From the last six Farm And Ranch Irrigation Surveys



■ SIS (total reporting)

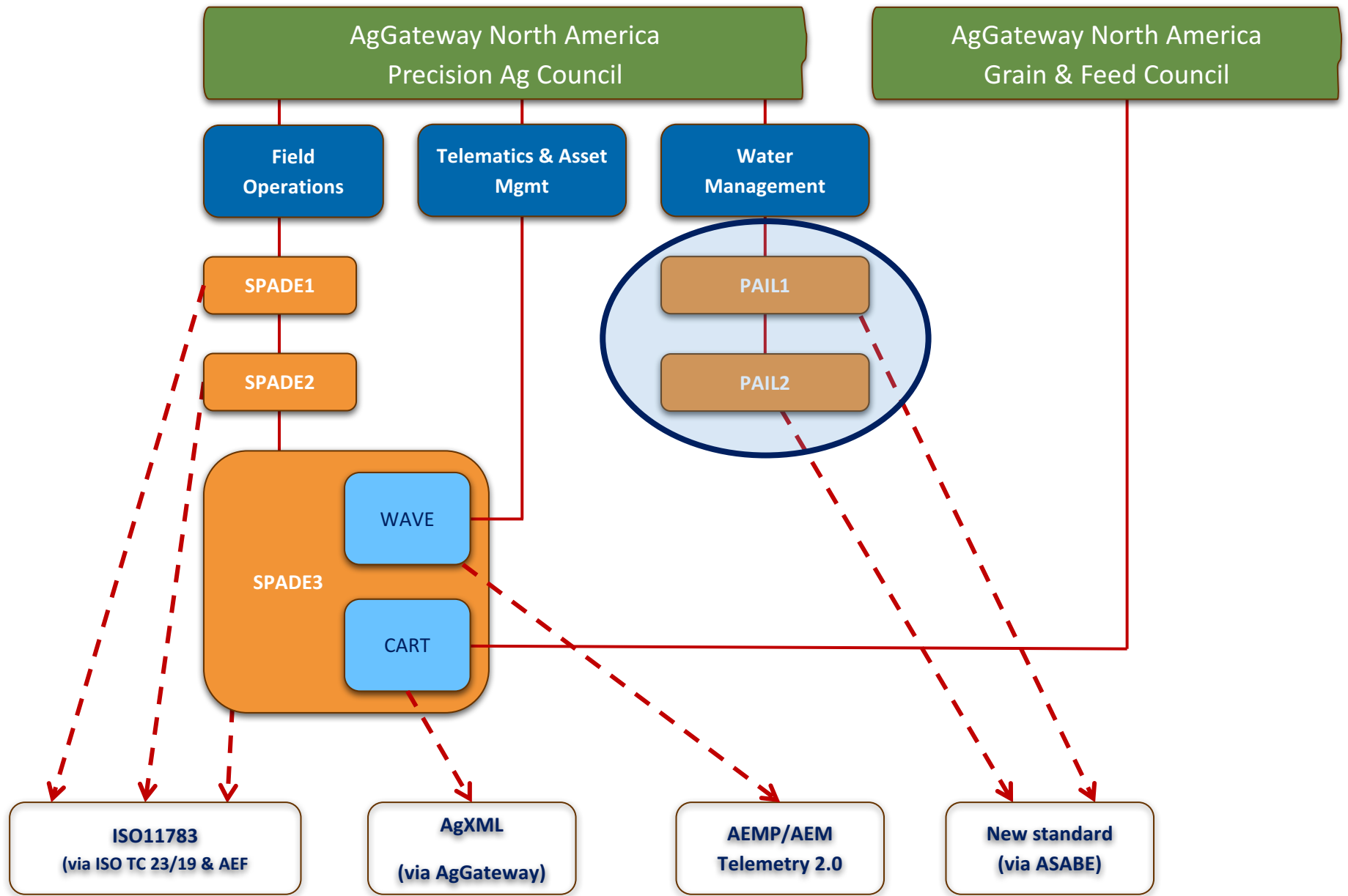
■ Non-SIS (total reporting)

The Interoperability Problem

- Scientific irrigation scheduling is infrequently used by growers.
- Reason: Integrating the necessary data sources is very complex and time-consuming.
- Northwestern Energy Efficiency Alliance (NEEA) concluded that a common data format for irrigation decision-making inputs and outputs would be a major first step toward greater interoperability.

AgGateway

- Nonprofit consortium of 240+ members
- Mission: Promote, enable and expand eAgriculture.
 - Strong emphasis on **implementing existing standards**
 - Strong emphasis on **collaboration**
- Membership
 - Open; members are primarily **businesses**.
 - Other **organizations** typically join as Associate members
 - There is a category for **individual** memberships.
- **Transparent** funding, governance, anti-trust and IP framework.
- Authority: *De facto* (Implementation by stakeholders)
 - Output often handed off to ISO, other *de jure* organizations.
- Expertise: **Supply chain and field operations business processes**
- Field operations interoperability initiatives: SPADE, PAIL, ADAPT.



PAIL Goal & Objective

- **Goal** : promote use of irrigation management technology by facilitating integration of disparate management systems
- **Objective**: provide an industry-wide format that will enable the exchange and use of data from irrigation management systems.

Scope

Observations



Operations



The Core Documents (to date)

- **Plan**

- "This is how we are going to grow this crop this season"

- **Observations and Measurements:**

- "This is happening out in the field"

- **Recommendation**

- "This is what I recommend we do about it"

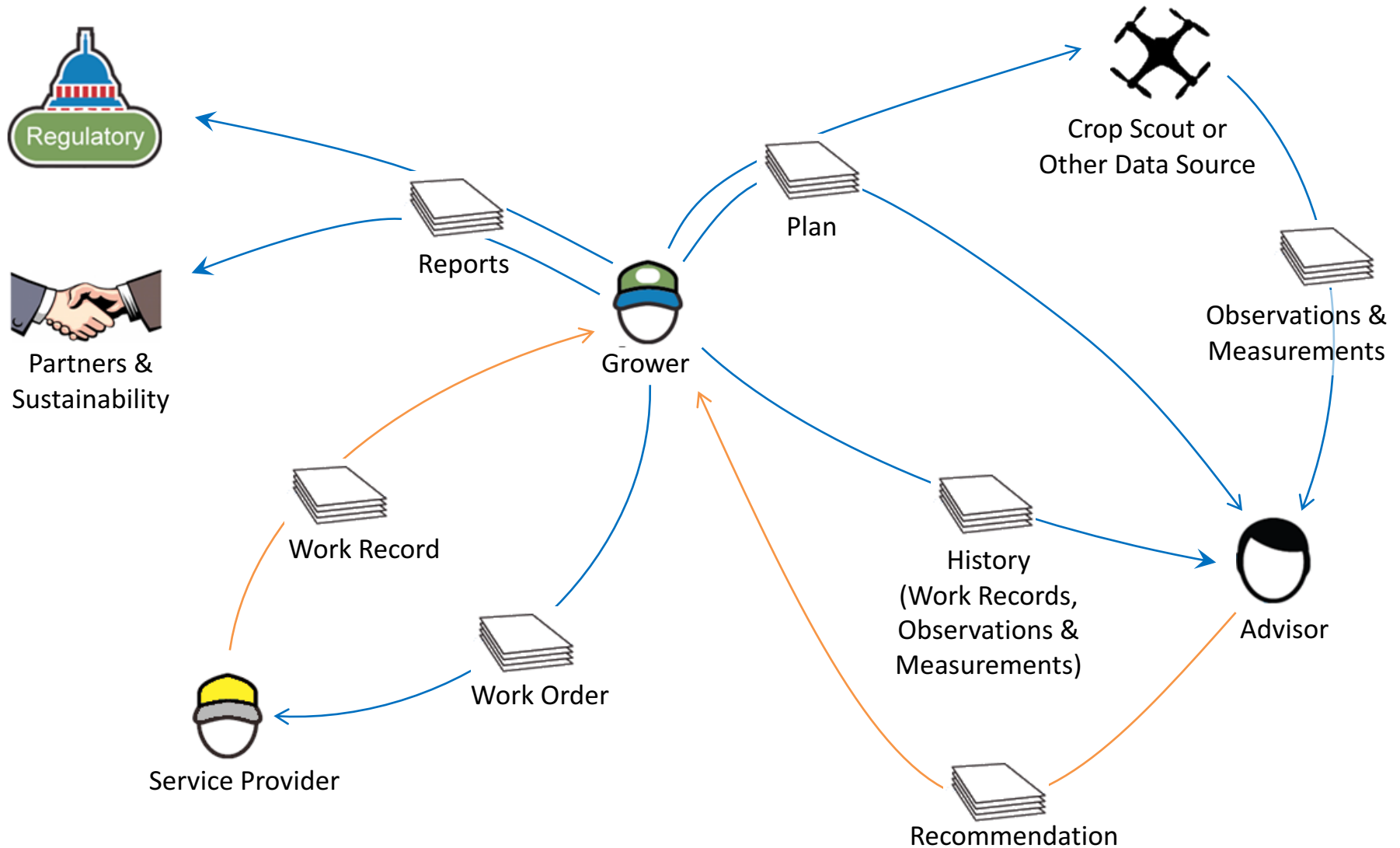
- **Work Order**

- "This is what we are going to do"

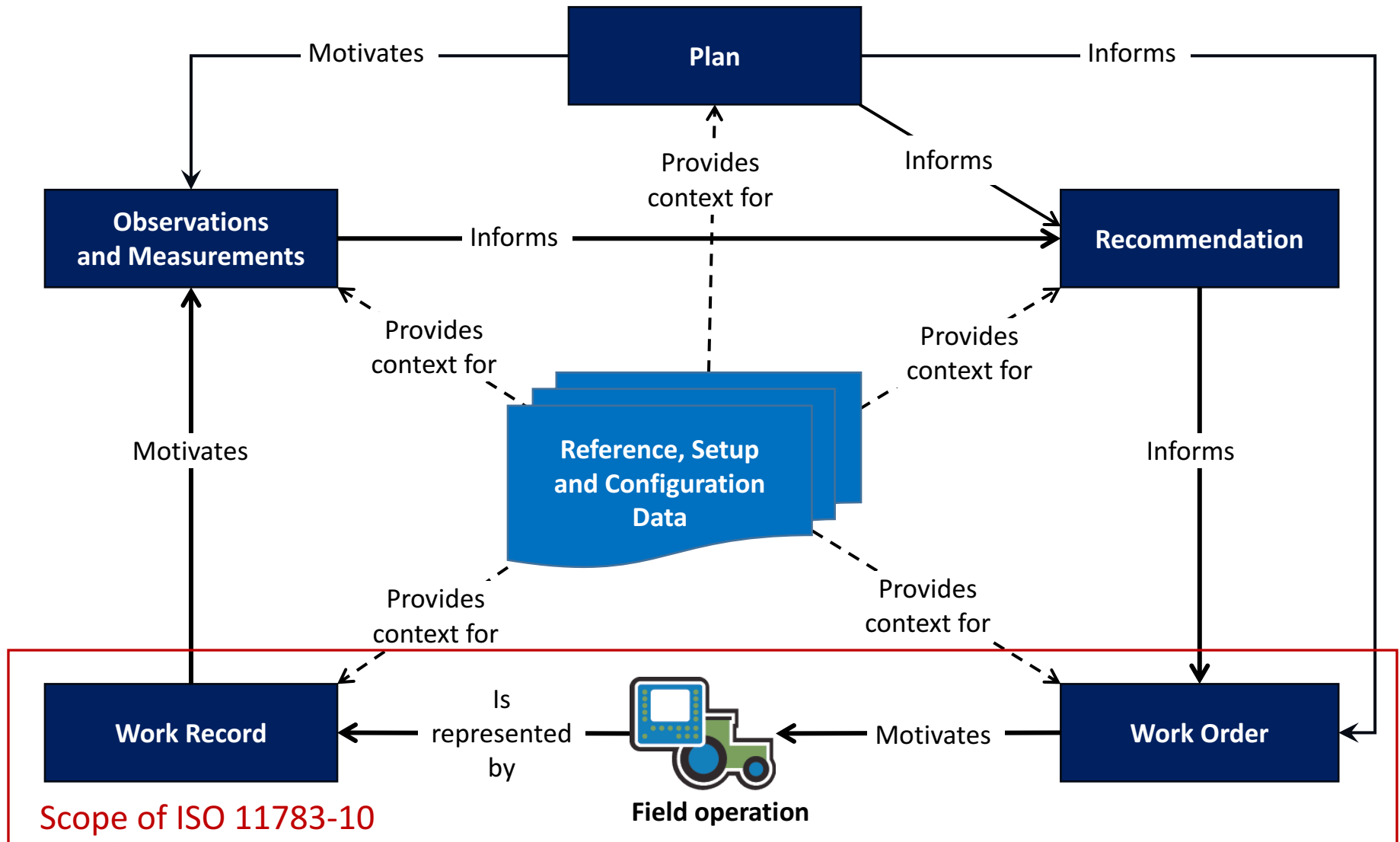
- **Work Record**

- "This is what we actually did"

Core Document Flow



Core Documents and their Relationships



Question	Description	Source
What	The products or services being applied, or the data being reported.	ISO 11783 + SPADE semantics
Where	Grower / Farms / Fields / Cropzones / GPS locations	ISO 11783 + SPADE + ISO 19112
Who	People involved and their roles: grower, operator, agronomist, trucker, customer, etc.	SPADE + Semantics
When	When should / did the operation happen?	ISO 6709 + SPADE semantics
How	Product rates, equipment settings, etc.	ISO 11783, 19115 , 19156, 19157 + SPADE
With What	What equipment is involved?	ISO 11783, 19115 , 19156, 19157 + Deere + ADAPT
Why	What was the reason for performing the operation?	SPADE semantics
Local data	Context items: A generic system to encode geopolitical-context-dependent information such as (for the US) FSA, EPA, DOT numbers, and so forth.	SPADE / ADAPT
Semantics	Infrastructure of variable-type registries	ISO 19135, 25964

Standards we are using or considering

- **Using**

- ISO 11783 (TC23/SC19): Farm machine control, data generation
- ISO 19112 (TC211): Lists of identified locations
- ISO 19156 (TC211): Observations and Measurements

- **In Progress / learning**

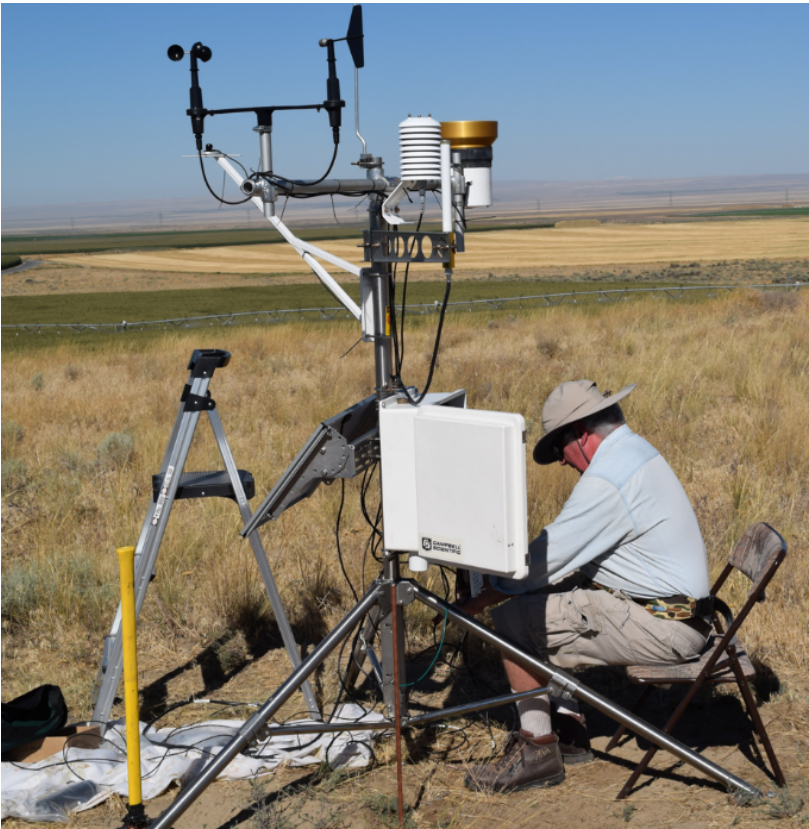
- ISO 19135 (TC211): Managing controlled vocabulary registries
- ISO 29564 (TC46/SC9/WG8): Thesauri

- **Considering / need starting point**

- ISO 19115 (TC211): Metadata
- ISO 19157 (TC211): Data Quality
- Sensor Observation Service
- SensorThings

PAIL Status

- Field Trail / Beta Test concluded this year and last
- Beta Test validated scope and depth of schema
- Submit draft standard in next few months



ASABE X632 Project

- ASABE Standards project
 - Parent committee is NRES-24
- Deliverable: US National Standard
- Coordinated with NRES-03/2 US TAG ISO TC23/SC18
- New, ad-hoc committee formed to guide movement through standards process
- Will be submitted to ISO TC23 / SC18 for consideration as a new standard after adoption by ASABE
- First standard in NRES-24 that contains an XML Schema
- Multipart standard similar to other ISO
 - Part 1: Common Elements
 - Part 2: Observations
 - Part 3: Operations
- Future Parts
 - Pumping & Flow Control
 - Drip/Micro Irrigation
 - Compliance testing
 - Chemigation / Fertigation

AgGateway / PAIL Stakeholders

AgGateway:
About 240
companies

Precision
Ag Council:
About 120
companies

PAIL Project:
20+ companies
(See below)



Questions?

(Including how you can participate)

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In-Season Management BPMN

